

SHOPLIST

APP LIST PARA SUAS COMPRAS



ShopList

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01

**Apresentação do
ShopList**

02

**Desenvolvimento/
código (principais
estruturas)**

03

**Funcionamento do
App**

O ShopList é um aplicativo Android desenvolvido em Java no Android Studio. Ele permite cadastrar produtos em uma lista de compras, informando nome, quantidade, unidade, preço e local de compra. O sistema também calcula automaticamente o total de itens e o valor total da compra.



ANDROIDMANIFEST.XML

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-permission android:name="android.permission.INTERNET" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@drawable/shoplist"
        android:label="@string/app_name"
        android:roundIcon="@drawable/shoplist"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppGerencia">

        <activity
            android:name=".SplashActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>

        <activity
            android:name=".MainActivity"
            android:exported="false" />

    </application>

</manifest>
```

Esse arquivo registra as telas do aplicativo e define qual tela será aberta primeiro.

Também define:

- Ícone
- Nome
- Tema visual do app



SPLASHACTIVITY.JAVA

```
package com.example.appgerencia;

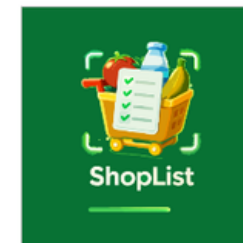
import android.content.Intent;
import android.os.Bundle;
import android.os.Handler;

import androidx.appcompat.app.AppCompatActivity;

public class SplashActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_splash);

        // Tela de carregamento rápida (1.5 segundos)
        new Handler().postDelayed(() -> {
            startActivity(new Intent(SplashActivity.this, MainActivity.class));
            finish();
        }, 1500);
    }
}
```



Pensado como uma
“Tela de carregamento”

Esse arquivo é a tela inicial do aplicativo que é apresentada por 1.5 segundos, antes de abrir a tela principal.

SHOPPINGITEM.JAVA

```
package com.example.appgerencia;

import java.io.Serializable;

public class ShoppingItem implements Serializable {
    private String name;
    private double quantity;
    private String unit;
    private double price;
    private String location;

    public ShoppingItem(String name, double quantity, String unit, double price, String location) {
        this.name = name;
        this.quantity = quantity;
        this.unit = unit;
        this.price = price;
        this.location = location;
    }

    public String getName() { return name; }
    public void setName(String name) { this.name = name; }

    public double getQuantity() { return quantity; }
    public void setQuantity(double quantity) { this.quantity = quantity; }

    public String getUnit() { return unit; }
    public void setUnit(String unit) { this.unit = unit; }

    public double getPrice() { return price; }
    public void setPrice(double price) { this.price = price; }

    public String getLocation() { return location; }
    public void setLocation(String location) { this.location = location; }
}
```

Representa o produto da lista de compras, criação da classe e seus atributos.

DIALOG_SHOPPING_ITEM.XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/editTextName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Nome do Produto"
        android:inputType="textCapSentences" />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:orientation="horizontal">

        <EditText
            android:id="@+id/editTextQuantity"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:hint="Qtd"
            android:inputType="numberDecimal" />

        <Spinner
            android:id="@+id/spinnerUnit"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1.5"
            android:layout_marginStart="8dp" />

    </LinearLayout>

    <EditText
        android:id="@+id/editTextPrice"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:hint="Preço (R$)"
        android:inputType="numberDecimal" />

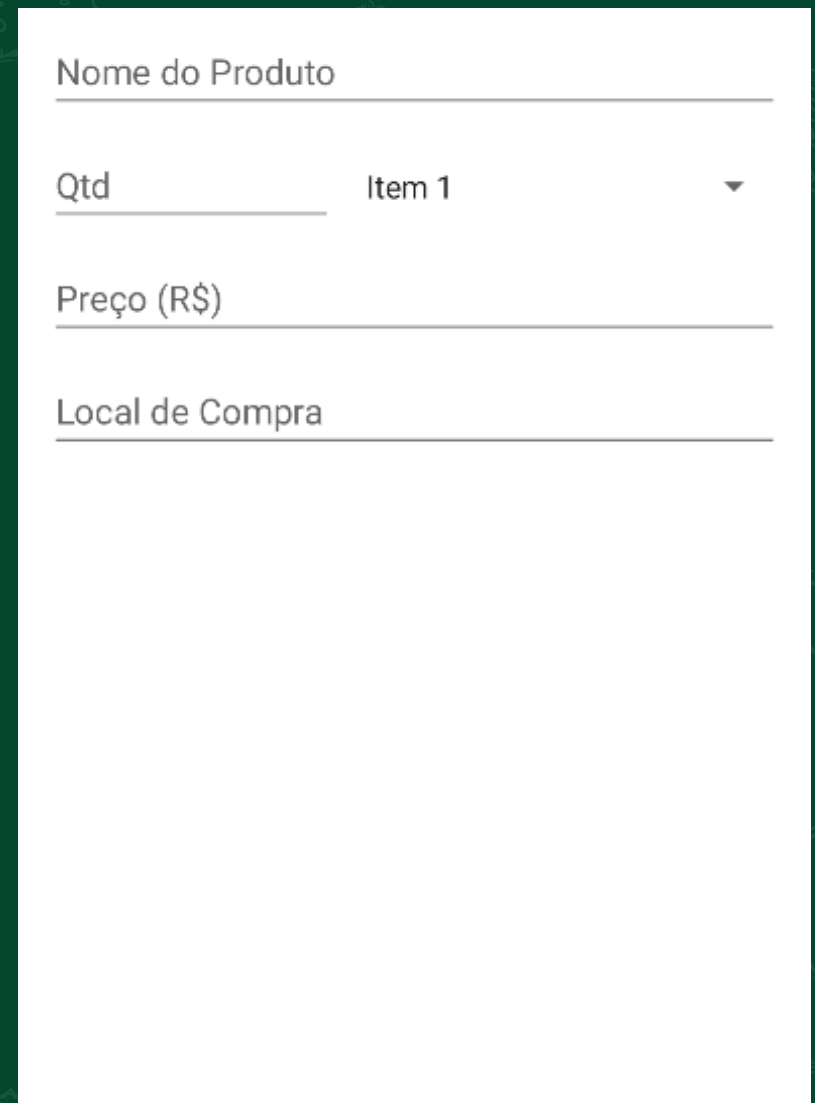
    <EditText
        android:id="@+id/editTextLocation"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="12dp"
        android:hint="Local de Compra"
        android:inputType="textCapWords" />

</LinearLayout>
```

```
<EditText
    android:id="@+id/editTextPrice"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="12dp"
    android:hint="Preço (R$)"
    android:inputType="numberDecimal" />

<EditText
    android:id="@+id/editTextLocation"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="12dp"
    android:hint="Local de Compra"
    android:inputType="textCapWords" />

</LinearLayout>
```



A visual representation of the XML layout, showing a white dialog box with rounded corners. It contains four input fields: a text field for 'Nome do Produto', a quantity field labeled 'Qtd' with a dropdown menu showing 'Item 1', a text field for 'Preço (R\$)', and a text field for 'Local de Compra'.

Esse arquivo representa o formulário usado para cadastro ou edição de produtos

ITEM_SHOPPING.XML

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.cardview.widget.CardView xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="8dp"
    app:cardCornerRadius="12dp"
    app:cardElevation="6dp">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:padding="16dp"
        android:background="@color/itemCardBG"
        android:gravity="center_vertical">

        <LinearLayout
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:orientation="vertical">

            <TextView
                android:id="@+id/textViewName"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:textSize="18sp"
                android:textStyle="bold"
                android:textColor="@color/primaryGreen"
                android:text="Nome do Produto" />

            <LinearLayout
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:orientation="horizontal"
                android:layout_marginTop="4dp">
```

```
                <TextView
                    android:id="@+id/textViewQuantity"
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:textSize="14sp"
                    android:textColor="#5D4037"
                    android:text="2 pacotes" />

                <TextView
                    android:id="@+id/textViewPrice"
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:layout_marginStart="16dp"
                    android:textSize="14sp"
                    android:textColor="@color/accentRed"
                    android:textStyle="bold"
                    android:text="R$ 0,00" />
            </LinearLayout>

            <TextView
                android:id="@+id/textViewLocation"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:layout_marginTop="2dp"
                android:textSize="12sp"
                android:textColor="#757575"
                android:text="Local: Supermercado" />
        </LinearLayout>

        <ImageButton
            android:id="@+id/buttonEdit"
            android:layout_width="40dp"
            android:layout_height="40dp"
            android:background="?attr/selectableItemBackgroundBorderless"
            android:contentDescription="Editar"
            android:src="@android:drawable/ic_menu_edit"
            app:tint="@color/primaryGreen" />

        <ImageButton
            android:id="@+id/buttonDelete"
            android:layout_width="40dp"
            android:layout_height="40dp"
            android:background="?attr/selectableItemBackgroundBorderless"
            android:contentDescription="Excluir"
            android:src="@android:drawable/ic_menu_delete"
            app:tint="@color/accentRed" />
    </LinearLayout>
</androidx.cardview.widget.CardView>
```

Nome do Produto

2 pacotes **R\$ 0,00**

Local: Supermercado



Esse arquivo define como cada produto e suas respectivas informações aparecerão na lista,

SHOPPINGADAPTER.JAVA

```
package com.example.appgerencia;

import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ImageButton;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

import java.util.List;
import java.util.Locale;

public class ShoppingAdapter extends RecyclerView.Adapter<ShoppingAdapter.ShoppingViewHolder> {

    private List<ShoppingItem> itemList;
    private OnItemClickListener listener;

    public interface OnItemClickListener {
        void onItemClick(int position);
        void onDeleteClick(int position);
    }

    public ShoppingAdapter(List<ShoppingItem> itemList, OnItemClickListener listener) {
        this.itemList = itemList;
        this.listener = listener;
    }

    @NonNull
    @Override
    public ShoppingViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int viewType) {
        View view = LayoutInflater.from(parent.getContext()).inflate(R.layout.item_shopping, parent, false);
        return new ShoppingViewHolder(view);
    }
```

```
    @Override
    public void onBindViewHolder(@NonNull ShoppingViewHolder holder, int position) {
        ShoppingItem item = itemList.get(position);
        holder.textViewName.setText(item.getName());

        String qtyText = String.format(Locale.getDefault(), "%.2f %s", item.getQuantity(), item.getUnit());
        holder.textViewQuantity.setText(qtyText);

        holder.textViewPrice.setText(String.format(Locale.getDefault(), "R$ %.2f", item.getPrice()));
        holder.textViewLocation.setText("Local: " + item.getLocation());

        holder.buttonEdit.setOnClickListener(v -> listener.onItemClick(position));
        holder.buttonDelete.setOnClickListener(v -> listener.onDeleteClick(position));
        holder.itemView.setOnClickListener(v -> listener.onItemClick(position));
    }

    @Override
    public int getItemCount() {
        return itemList.size();
    }

    public static class ShoppingViewHolder extends RecyclerView.ViewHolder {
        TextView textViewName, textViewQuantity, textViewPrice, textViewLocation;
        ImageButton buttonEdit, buttonDelete;

        public ShoppingViewHolder(@NonNull View itemView) {
            super(itemView);
            textViewName = itemView.findViewById(R.id.textViewName);
            textViewQuantity = itemView.findViewById(R.id.textViewQuantity);
            textViewPrice = itemView.findViewById(R.id.textViewPrice);
            textViewLocation = itemView.findViewById(R.id.textViewLocation);
            buttonEdit = itemView.findViewById(R.id.buttonEdit);
            buttonDelete = itemView.findViewById(R.id.buttonDelete);
        }
    }
}
```

Esse arquivo tem o papel de transformar cada SoppingItem em um card visual

MAINACTIVITY.JAVA

```
package com.example.appgerencia;

import android.content.SharedPreferences;
import android.os.Bundle;
import android.text.Editable;
import android.text.TextWatcher;
import android.view.LayoutInflater;
import android.view.View;
import android.widget.AdapterView;
import android.widget.AdapterView.OnItemClickListener;
import android.widget.EditText;
import android.widget.Spinner;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import com.google.android.material.floatingactionbutton.FloatingActionButton;
import com.google.gson.Gson;
import com.google.gson.reflect.TypeToken;

import java.lang.reflect.Type;
import java.text.NumberFormat;
import java.util.ArrayList;
import java.util.List;
import java.util.Locale;

import coil.Coil;
import coil.request.CachePolicy;
import coil.request.ImageRequest;
```

```
public class MainActivity extends AppCompatActivity implements ShoppingAdapter.OnItemClickListener {

    private List<ShoppingItem> shoppingList;
    private ShoppingAdapter adapter;
    private RecyclerView recyclerView;
    private FloatingActionButton fabAdd;
    private TextView textViewTotalItems, textViewTotalPrice;
    private final String PREFS_NAME = "ShopListPrefs";
    private final String LIST_KEY = "shopping_list";
    private final Locale ptBr = new Locale("pt", "BR");

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        loadData();
        preloadImages();

        recyclerView = findViewById(R.id.recyclerView);
        textViewTotalItems = findViewById(R.id.textViewTotalItems);
        textViewTotalPrice = findViewById(R.id.textViewTotalPrice);

        recyclerView.setLayoutManager(new LinearLayoutManager(this));
        adapter = new ShoppingAdapter(shoppingList, this);
        recyclerView.setAdapter(adapter);

        fabAdd = findViewById(R.id.fabAdd);
        fabAdd.setOnClickListener(v -> showItemDialog(null, -1));

        updateTotals();
    }
}
```

Representa a classe principal do app

- Import (s)
- public class MainActivity
- onCreate

MAINACTIVITY.JAVA

```
private void preloadImages() {
    if (shoppingList == null) return;
    for (ShoppingItem item : shoppingList) {
        String cleanName = item.getName().toLowerCase().trim().replace(" ", ",");
        int lockId = Math.abs(item.getName().toLowerCase().trim().hashCode() % 1000);
        String imageUrl = "https://loremflickr.com/300/300/" + cleanName + ",grocery/all?lock=" + lockId;

        ImageRequest request = new ImageRequest.Builder(this)
            .data(imageUrl)
            .diskCachePolicy(CachePolicy.ENABLED)
            .build();
        Coil.imageLoader(this).enqueue(request);
    }
}

private void updateTotals() {
    double totalItems = 0;
    double totalPrice = 0;
    for (ShoppingItem item : shoppingList) {
        totalItems += item.getQuantity();
        totalPrice += (item.getQuantity() * item.getPrice());
    }
    textViewTotalItems.setText(String.format(Locale.getDefault(), "Total de Itens: %.2f", totalItems));
    textViewTotalPrice.setText(NumberFormat.getCurrencyInstance(ptBr).format(totalPrice));
}
```

```
private void showItemDialog(ShoppingItem item, int position) {
    AlertDialog.Builder builder = new AlertDialog.Builder(this);
    LayoutInflater inflater = getLayoutInflater();
    View dialogView = inflater.inflate(R.layout.dialog_shopping_item, null);
    builder.setView(dialogView);

    EditText editTextName = dialogView.findViewById(R.id.editTextName);
    EditText editTextQuantity = dialogView.findViewById(R.id.editTextQuantity);
    Spinner spinnerUnit = dialogView.findViewById(R.id.spinnerUnit);
    EditText editTextPrice = dialogView.findViewById(R.id.editTextPrice);
    EditText editTextLocation = dialogView.findViewById(R.id.editTextLocation);

    String[] units = {"pacotes", "kilos", "gramas", "unidades", "latas", "garrafas", "barras", "outros"};
    ArrayAdapter<String> unitAdapter = new ArrayAdapter<>(this, android.R.layout.simple_spinner_item, units);
    unitAdapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
    spinnerUnit.setAdapter(unitAdapter);

    addCapitalizeWatcher(editTextName);
    addCapitalizeWatcher(editTextLocation);

    editTextPrice.addTextChangedListener(new TextWatcher() {
        private String current = "";
        @Override
        public void beforeTextChanged(CharSequence s, int start, int count, int after) {}
        @Override
        public void onTextChanged(CharSequence s, int start, int before, int count) {
            if (!s.toString().equals(current)) {
                editTextPrice.removeTextChangedListener(this);
                String cleanString = s.toString().replaceAll("[^\\d]", "");
                if (cleanString.isEmpty()) cleanString = "0";
                double parsed = Double.parseDouble(cleanString);
                String formatted = NumberFormat.getCurrencyInstance(ptBr).format((parsed / 100));
                current = formatted;
                editTextPrice.setText(formatted);
                editTextPrice.setSelection(formatted.length());
                editTextPrice.addTextChangedListener(this);
            }
        }
        @Override
        public void afterTextChanged(Editable s) {}
    });
}
```

- preloadImages
- updateTotals
- showItemDialog
- editTextPrice

MAINACTIVITY.JAVA

- set (s)
- addCapitalizeWatcher
- capitalizeWords

```
if (item != null) {
    builder.setTitle("Editar Produto");
    editTextName.setText(item.getName());
    editTextQuantity.setText(String.valueOf(item.getQuantity()));
    long priceInCents = Math.round(item.getPrice() * 100);
    editTextPrice.setText(String.valueOf(priceInCents));
    editTextLocation.setText(item.getLocation());
    for (int i = 0; i < units.length; i++) {
        if (units[i].equals(item.getUnit())) {
            spinnerUnit.setSelection(i);
            break;
        }
    }
} else {
    builder.setTitle("Novo Produto");
    editTextPrice.setText("0");
}

builder.setPositiveButton("Salvar", (dialog, which) -> {
    String name = editTextName.getText().toString().trim();
    String loc = editTextLocation.getText().toString().trim();
    String qtyStr = editTextQuantity.getText().toString().replace(",", ".").trim();
    String priceStr = editTextPrice.getText().toString()
        .replaceAll("[R$\\s]", "").replace(".", "").replace(",", ".");

    if (!name.isEmpty() && !qtyStr.isEmpty()) {
        try {
            double qty = Double.parseDouble(qtyStr);
            double price = Double.parseDouble(priceStr);
            String unit = spinnerUnit.getSelectedItem().toString();

            if (item == null) {
                shoppingList.add(new ShoppingItem(name, qty, unit, price, loc));
                adapter.notifyItemInserted(shoppingList.size() - 1);
            } else {
                item.setName(name);
                item.setQuantity(qty);
                item.setUnit(unit);
                item.setPrice(price);
                item.setLocation(loc);
                adapter.notifyItemChanged(position);
            }

            saveData();
            updateTotals();
            preloadImages();
        } catch (NumberFormatException e) {
            Toast.makeText(this, "Erro nos valores", Toast.LENGTH_SHORT).show();
        }
    }
});
builder.setNegativeButton("Cancelar", null);
builder.create().show();
}
```

```
private void addCapitalizeWatcher(EditText editText) {
    editText.addTextChangedListener(new TextWatcher() {
        private boolean isUpdating = false;
        @Override
        public void beforeTextChanged(CharSequence s, int start, int count, int after) {}
        @Override
        public void onTextChanged(CharSequence s, int start, int before, int count) {
            if (isUpdating) return;
            String original = s.toString();
            String capitalized = capitalizeWords(original);
            if (!original.equals(capitalized)) {
                isUpdating = true;
                int selection = editText.getSelectionStart();
                editText.setText(capitalized);
                if (selection <= capitalized.length()) editText.setSelection(selection);
                isUpdating = false;
            }
        }
        @Override
        public void afterTextChanged(Editable s) {}
    });
}

private String capitalizeWords(String input) {
    if (input == null || input.isEmpty()) return input;
    StringBuilder capitalized = new StringBuilder();
    boolean nextTitleCase = true;
    for (char c : input.toCharArray()) {
        if (Character.isSpaceChar(c)) { nextTitleCase = true; }
        else if (nextTitleCase) { c = Character.toUpperCase(c); nextTitleCase = false; }
        capitalized.append(c);
    }
    return capitalized.toString();
}

@Override
public void onItemClick(int position) { showItemDialog(shoppingList.get(position), position); }
```


MAINACTIVITY.JAVA

```
@Override
public void onDeleteClick(int position) {
    new AlertDialog.Builder(this)
        .setTitle("Remover Item")
        .setMessage("Deseja remover " + shoppingList.get(position).getName() + "?")
        .setPositiveButton("Sim", (dialog, which) -> {
            shoppingList.remove(position);
            adapter.notifyItemRemoved(position);
            adapter.notifyItemRangeChanged(position, shoppingList.size());
            saveData();
            updateTotals();
        }).setNegativeButton("Não", null).show();
}

private void saveData() {
    SharedPreferences sharedPreferences = getSharedPreferences(PREFS_NAME, MODE_PRIVATE);
    SharedPreferences.Editor editor = sharedPreferences.edit();
    Gson gson = new Gson();
    editor.putString(LIST_KEY, gson.toJson(shoppingList));
    editor.apply();
}

private void loadData() {
    SharedPreferences sharedPreferences = getSharedPreferences(PREFS_NAME, MODE_PRIVATE);
    String json = sharedPreferences.getString(LIST_KEY, null);
    Type type = new TypeToken<ArrayList<ShoppingItem>>().getType();
    shoppingList = new Gson().fromJson(json, type);
    if (shoppingList == null) {
        shoppingList = new ArrayList<>();
        shoppingList.add(new ShoppingItem("Coca Cola", 1.0, "unidades", 8.50, "Bebidas"));
        shoppingList.add(new ShoppingItem("Snickers", 1.0, "unidades", 5.50, "Doces"));
    }
}
```

- onDeleteClick
- saveData
- loadData

BUILD.GRADLE.KTS

```
plugins {  
    alias(libs.plugins.android.application)  
}  
  
android {  
    namespace = "com.example.appgerencia"  
    compileSdk = 35  
  
    defaultConfig {  
        applicationId = "com.example.appgerencia"  
        minSdk = 28  
        targetSdk = 35  
        versionCode = 1  
        versionName = "1.0"  
  
        testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"  
    }  
  
    buildTypes {  
        release {  
            isMinifyEnabled = false  
            proguardFiles(  
                getDefaultProguardFile("proguard-android-optimize.txt"),  
                "proguard-rules.pro"  
            )  
        }  
    }  
    compileOptions {  
        sourceCompatibility = JavaVersion.VERSION_11  
        targetCompatibility = JavaVersion.VERSION_11  
    }  
}  
  
dependencies {  
    implementation(libs.appcompat)  
    implementation(libs.material)  
    implementation(libs.activity)  
    implementation(libs.constraintlayout)  
    implementation(libs.gson)  
    implementation(libs.coil)  
    implementation(libs.coil.svg)  
    testImplementation(libs.junit)  
    androidTestImplementation(libs.ext.junit)  
    androidTestImplementation(libs.espresso.core)  
}
```

The background is a vibrant yellow-orange color, densely populated with small, white, hand-drawn food icons. These icons include various items such as slices of pizza, whole pizzas, apples, donuts, ice cream cones, chicken drumsticks, and pieces of cheese. In the center of the image, there is a dark green rounded rectangle containing the text '03' and 'Funcionamento do App'. Above the '03' is a small green circle. The bottom right corner of the image features a large, dark green curved shape that overlaps the yellow background.

03

Funcionamento do App

MUITO OBRIGADO!



ShopList